

# THERMAL IMAGING REPORT

Property: Greenfield Residency, Block C | Report No: BSI-THR-2026-047

Date: 28 March 2026

Inspector: John Matthews | Firm: BuildSafe Inspections Pvt. Ltd.

## Equipment & Methodology

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Camera Model:        | FLIR E86 Thermal Imaging Camera                  |
| Resolution:          | 464 x 348 IR pixels                              |
| Temperature Range:   | -20°C to +550°C                                  |
| Emissivity Setting:  | 0.95 (standard for building materials)           |
| Ambient Temperature: | 32°C                                             |
| Inspection Time:     | 6:00 AM — 8:30 AM (pre-dawn for accurate delta)  |
| Operator:            | Priya Sharma, Certified Thermographer (Level II) |
| Report Reference:    | BSI-THR-2026-047                                 |

Thermal imaging was conducted in the early morning to maximize temperature differential between wet/damaged areas and dry building fabric. All readings are surface temperature measurements. Anomalies are defined as areas showing greater than 2°C delta from surrounding baseline material.

## Thermal Finding 1 — Rooftop & Terrace

### Thermal Observations:

Thermal scanning of the rooftop membrane revealed large zones of moisture entrapment beneath the waterproofing layer. Wet areas retain heat longer than dry areas and appear as warm anomalies in pre-dawn scans.

| Zone              | Baseline Temp | Anomaly Temp | Delta  | Interpretation                      |
|-------------------|---------------|--------------|--------|-------------------------------------|
| North-west corner | 28.2°C        | 34.7°C       | +6.5°C | High moisture entrapment            |
| Central roof zone | 28.1°C        | 31.4°C       | +3.3°C | Moderate moisture beneath membrane  |
| Parapet wall base | 27.9°C        | 33.1°C       | +5.2°C | Water ingress at failed joint       |
| Water tank base   | 28.0°C        | 35.8°C       | +7.8°C | Active water leak — high confidence |

Thermal Image Reference: THR-047-RF-01 (North-west corner moisture map), THR-047-RF-02 (Water tank base active leak). Image Not Available in this document — refer to thermal image annexure.

## Thermal Finding 2 — East External Facade

### Thermal Observations:

Scanning of the east facade from ground level using a telescopic platform revealed cold anomalies around window frames consistent with air infiltration and moisture presence behind the external plaster.

| Location | Baseline Temp | Anomaly Temp | Delta | Interpretation |
|----------|---------------|--------------|-------|----------------|
|----------|---------------|--------------|-------|----------------|

|                                   |        |        |                                     |
|-----------------------------------|--------|--------|-------------------------------------|
| Floor 2 — Window frames (3 units) | 26.4°C | -2.7°C | Air infiltration / failed sealant   |
| Floor 3 — Window frames (3 units) | 25.8°C | -3.2°C | Significant moisture behind plaster |
| Floor 3 — Plaster detachment      | 24.1°C | -5.1°C | Delamination air gap confirmed      |
| Floor 2 — Below window sill       | 27.6°C | -1.4°C | Minor seepage — monitor             |

Thermal Image Reference: THR-047-EF-01 (Floor 3 plaster delamination), THR-047-EF-02 (Window frame cold zones).  
Image Not Available in this document — refer to thermal image annexure.

### Thermal Finding 3 — Third Floor Corridor Ceiling

**Thermal Observations:**

Interior ceiling scan of the third floor corridor confirmed active moisture migration from the roof slab above. The thermal pattern is consistent with water tracking along the construction joint between the roof slab and parapet wall.

- Ceiling surface temperature anomaly: baseline 29.5°C vs anomaly zone 34.2°C (+4.7°C) — active moisture.
- Warm patch extends approximately 6.2m in length along the north-west corridor section.
- Peak temperature recorded at 35.1°C directly above Unit C-302 doorway.
- Diagonal crack location shows +3.8°C compared to adjacent dry ceiling — moisture present in crack.
- Electrical conduit in affected zone shows surface temperature of 31.2°C vs baseline 28.8°C — heat buildup.

Thermal Image Reference: THR-047-C3-01 (Full corridor ceiling scan), THR-047-C3-02 (Crack zone close-up). Image Not Available in this document — refer to thermal image annexure.

### Thermal Finding 4 — Basement Parking Level

**Thermal Observations:**

Basement ceiling and north retaining wall were scanned. Active water seepage locations were confirmed via thermal differential. The retaining wall shows a broad moisture front consistent with groundwater ingress.

- Ceiling slab joint (Location 1): anomaly temp 33.6°C vs baseline 27.8°C (+5.8°C) — active drip confirmed.
- Ceiling slab joint (Location 2): anomaly temp 32.1°C vs baseline 27.9°C (+4.2°C) — active seepage.
- North retaining wall: broad cold anomaly -3.1°C delta over 4m x 2m zone — groundwater ingress.
- Sump pit area: surface temp elevated at 30.4°C vs ambient 28.0°C — pump motor heat detected.
- No significant thermal anomalies detected on south or west walls of basement.

Thermal Image Reference: THR-047-BM-01 (Ceiling joint moisture map), THR-047-BM-02 (Retaining wall cold front). Image Not Available in this document — refer to thermal image annexure.

### Thermal Finding 5 — Electrical Panel Scanning

**Thermal Observations:**

All three electrical distribution panels were scanned with the panel doors open under normal load conditions. Thermal anomalies in electrical components are indicative of loose connections, overloaded circuits, or failing components.

| Panel Location | Component               | Temp Reading | Safe Limit | Risk Level             |
|----------------|-------------------------|--------------|------------|------------------------|
| Ground Floor   | MCB Switch 3 & 4        | 87.4°C       | 60°C       | CRITICAL               |
| Ground Floor   | Feeder cable entry      | 74.2°C       | 60°C       | HIGH                   |
| Second Floor   | MCB Switch 2            | 68.1°C       | 60°C       | HIGH                   |
| Second Floor   | Bus bar connection      | 71.3°C       | 60°C       | HIGH                   |
| Fourth Floor   | Panel enclosure surface | 41.2°C       | 35°C       | MEDIUM — moisture risk |

Ground floor MCB temperatures at 87.4°C are 45% above safe operating limits and represent an imminent fire risk. Immediate shutdown and replacement recommended.

Thermal Image Reference: THR-047-EP-01 (Ground floor panel full scan), THR-047-EP-02 (MCB close-up hot spot). Image Not Available in this document — refer to thermal image annexure.

## Thermographer Summary & Conclusions

This thermal imaging survey has identified and quantified moisture and heat anomalies across five key areas of Block C, Greenfield Residency. The findings strongly corroborate the visual inspection observations and provide additional depth on the severity and extent of each issue.

### Priority Summary:

| Area                 | Thermal Severity | Key Finding                              |
|----------------------|------------------|------------------------------------------|
| Electrical Panels    | CRITICAL         | MCBs at 87.4°C — fire risk               |
| Third Floor Corridor | CRITICAL         | Active moisture +4.7°C above baseline    |
| Rooftop / Terrace    | HIGH             | Water tank leak +7.8°C anomaly           |
| Basement             | HIGH             | Two active ceiling drip points confirmed |
| East Facade          | MEDIUM           | Plaster delamination air gap confirmed   |

All thermal readings are surface temperatures measured under stated ambient conditions. This report should be read in conjunction with the visual inspection report BSI-2026-047. Thermographer: Priya Sharma (Level II Certified) | Equipment calibrated: January 2026.